

Knowledge Graph e CPS in Sanità

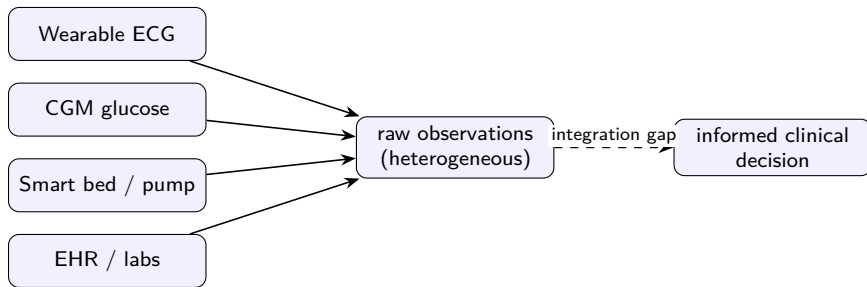
PhD Lecture Module

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From a Flood of Signals to a Single Decision



- Healthcare now runs on **streams of device data** — but a clinician needs **one coherent picture**, not four export files.
- This section answers: *why is integrating healthcare data hard, and why is a **knowledge graph (KG)** the right tool for it?*
- **Learning outcome:** by the end, you can justify a KG over a relational database for a concrete clinical integration task.

Roadmap & Time Budget

Section	Focus	Time
1. The practical problem	Heterogeneous device data & the integration gap	8 min
2. Knowledge Graph — the solution	Definition, pillars, mini-KG, vs. databases	10 min
3. Clinical relationships	Complexity & relational patterns	5 min
4. Healthcare ontologies	Why we reuse controlled vocabularies	2 min

- We move deliberately **from a concrete clinical problem → to a formal solution** — jargon is introduced only when it earns its place.
- The running example is a **diabetic patient** monitored by a CGM, a wearable ECG, and an EHR.

Discussion Prompt for PhD Students

Before we start: predict which of the four sections hides the *conceptual* core you would need for your own thesis. Note it down — we check at the end whether your intuition changed

1 · The Practical Problem

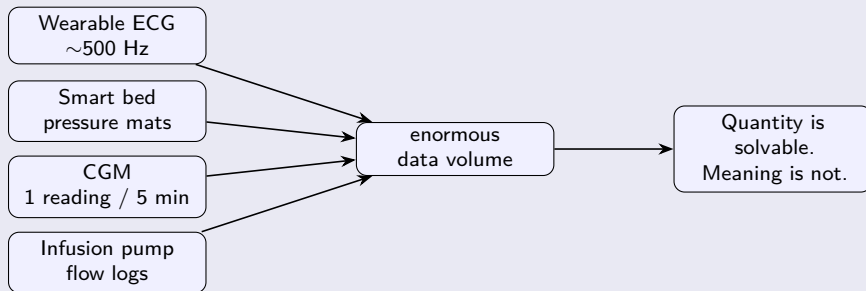
8 minutes — “*Devices speak; clinicians need sentences.*”

- Modern care is **device-saturated**: the question is no longer *getting* data, but **making it jointly meaningful**.
- We will isolate **four independent dimensions** in which healthcare data is heterogeneous — each one alone can break integration.

i Discussion Prompt for PhD Students

Name a clinical decision that is impossible to make correctly if you see only **one** device's data in isolation. (Hint: think drug–device interactions.)

The Data Deluge: Volume Is the Easy Part

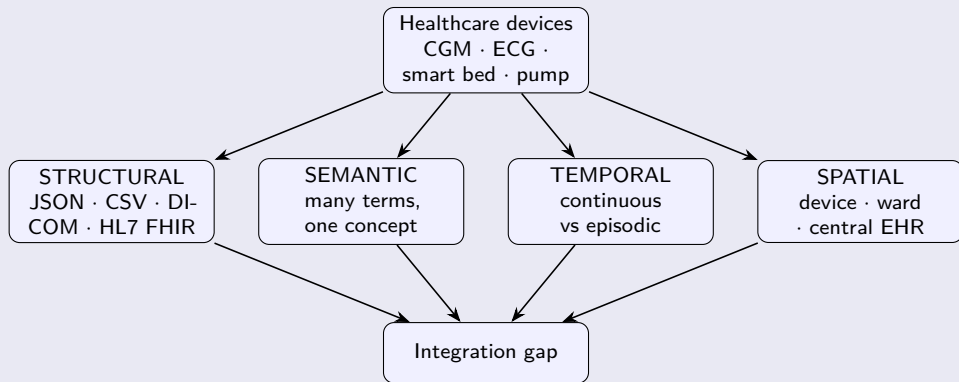


- A single ICU bed can generate **thousands of data points per second** across ECG waveforms, ventilator telemetry, and infusion logs.
- **Volume alone is an engineering problem** (storage, throughput) — and it is largely solved.
- The unsolved problem is **heterogeneity**: the data was never designed to be read together.

i Discussion Prompt for PhD Students

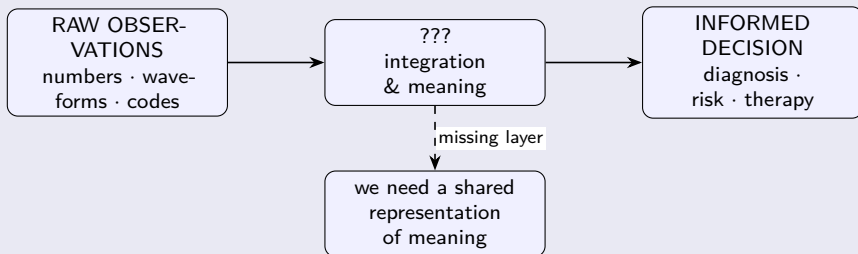
Storage is cheap. So why does “having all the data” still not give you the answer? What exactly is missing between the raw byte and the clinical insight?

Four Dimensions of Heterogeneity



- **Structural** — each device/export uses a different *format* (a wearable ships JSON, imaging ships DICOM, the EHR speaks HL7 FHIR).
- **Semantic** — the *same* clinical concept (e.g., “blood glucose”) is named differently across systems and terminologies.
- **Temporal** — sampling rates clash: a CGM every 5 min, an ECG continuously, a lab test once a day.
- **Spatial** — data lives in **different places**: a personal device, a ward monitor, a central EHR — each behind its own interface.

The Core Challenge: Observations → Decisions

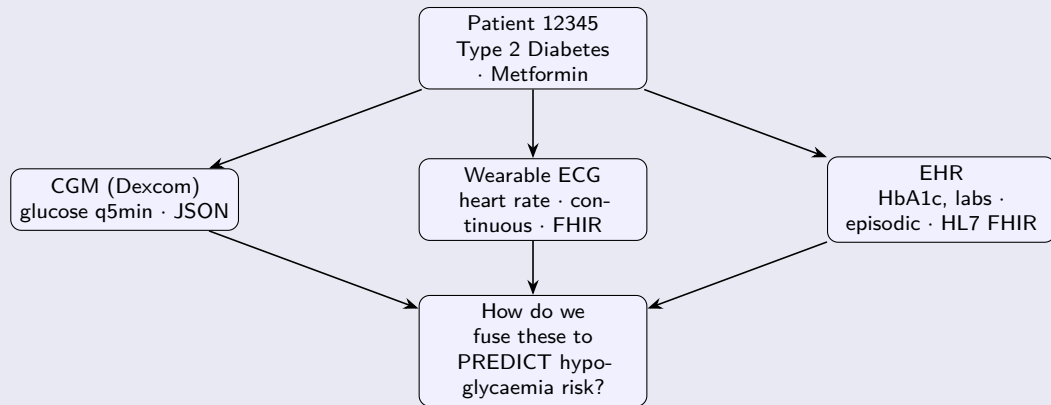


- A glucose reading, an ECG strip, and a lab result are **observations** — facts about a patient at an instant.
- A clinician needs a **decision**: *“Is this patient heading toward hypoglycaemia in the next hour?”*
- Between observation and decision sits a **missing layer**: a representation that says *what each value means, how they relate, and what they imply together*.

! The crux

The integration problem is **not** “join the tables.” It is “*agree on what the data means.*” A knowledge graph is a representation engineered for exactly that.

Worked Example: One Diabetic Patient, Three Data Worlds



- **CGM** streams glucose every 5 minutes; **ECG** records heart rate continuously; **EHR** holds an HbA1c from last week's visit.
- Three sources, three formats, three time scales, three locations — describing **one patient**.
- **Clinical question:** *can we predict an impending hypoglycaemic event and warn before it happens?*
- Answering this requires the sources to **agree on meaning and align in time** — the integration gap made concrete.

2 · The Knowledge Graph — The Solution

10 minutes — *“A graph that gives data its meaning.”*

- We now formalise the **missing layer** from Section 1 and give it a name: the **knowledge graph (KG)**.
- Definition → three pillars → a worked mini-KG → why it beats a plain database.

i Discussion Prompt for PhD Students

Before the formal definition: in your own words, what distinguishes a “knowledge graph” from “just a diagram with boxes and arrows”? Hold that intuition — we will sharpen it.

What Is a Knowledge Graph? (Formal Definition)

 Definition (after Hogan et al., 2021)

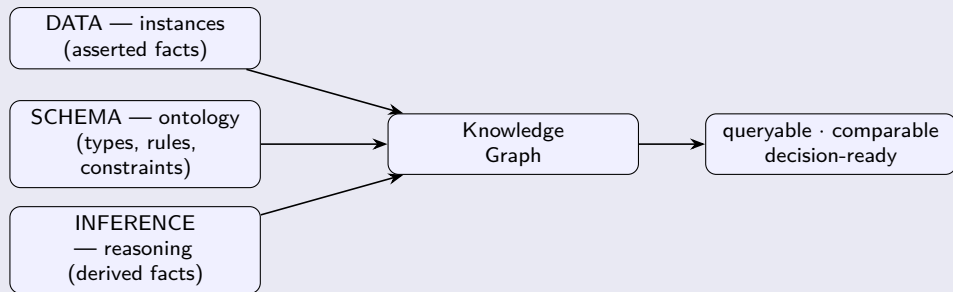
A **knowledge graph** is a **directed, labelled graph** in which **nodes represent entities** and **edges represent named semantic relationships** between them, backed by a **formal schema (ontology)** that gives those relationships a well-defined meaning and supports **automated reasoning**.

- **Directed** — edges have a direction (Patient $\rightarrow$ hasDiagnosis $\rightarrow$ Diagnosis), because *A relates to B* does not imply *B relates to A*.
- **Labelled** — every edge carries a **type** (hasDiagnosis, assumes, monitoredBy) that is itself a first-class concept.
- **Semantically grounded** — relationships are not arbitrary strings; they are defined in an **ontology** so machines can interpret them consistently.

 Intuition

If a spreadsheet stores *values*, a KG stores **statements of the form *subject–predicate–object*** — i.e., *who is related to whom, and how*.

The Three Pillars of a Knowledge Graph

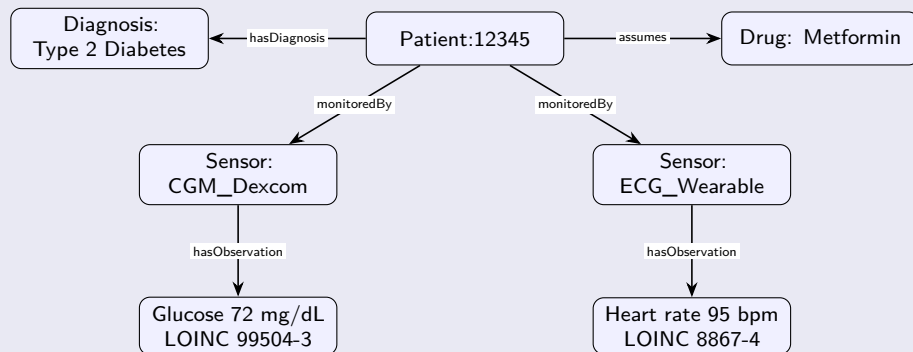


- **Data (A-box / instances)** — the concrete facts: *Patient 12345 takes Metformin.*
- **Schema (T-box / ontology)** — the rules and types: *Metformin is a Drug; a Patient may assume a Drug.*
- **Inference (reasoning)** — new facts the engine **derives** rather than stores: *because Metformin is a biguanide, and biguanides carry lactic-acidosis risk under contrast, the patient is flagged.*

i Discussion Prompt for PhD Students

A relational database has *data* and *schema*. Which of the three pillars does it **completely lack** — and why is that pillar the clinically dangerous one to be missing?

Mini-KG for Our Diabetic Patient



- **Nodes (entities):** Patient:12345, Diagnosis:Type2Diabetes, Drug:Metformin, Sensor:CGM_Dexcom, Sensor:ECG_Wearable.
- **Edges (relationships):** hasDiagnosis, assumes, monitoredBy, hasObservation.
- Every node and edge carries a **standard code** (LOINC for observations) — so another hospital reads the *same meaning*, not just the same text.
- One graph now holds **the patient, their disease, their therapy, and their live sensor observations** together.

How the KG Solves the Integration Problem

- **Standardises terminology** — diagnoses map to **SNOMED CT**, observations to **LOINC**; “glucose”, “Glc”, and “blood sugar” collapse to one concept.
- **Links disparate entities into a single representation** — CGM, ECG and EHR become *one connected graph* about one patient, regardless of source format.
- **Enables inference** — the engine derives facts no one explicitly recorded.

💡 Inference in action

Patient assumes Metformin + patient must undergo contrast-enhanced CT → risk of lactic acidosis — flagged automatically, without a clinician coding the check by hand.

- The three Section-1 heterogeneities (**structural, semantic, temporal**) are each absorbed by a different KG mechanism: ontology mapping, shared codes, and time-stamped observations.

i Discussion Prompt for PhD Students

The lactic-acidosis rule fires from two facts the KG never stored together. Whose responsibility is the *correctness* of that derived fact — the ontology author, the reasoning engine, or the clinician?

Why Not a Database? KG vs Traditional Approaches

Approach	Relationships	Schema flexibility	Inference
Relational DB	via expensive multi-table JOINS	rigid (schema-first)	none
Document store (JSON/NoSQL)	no explicit relationships	flexible but shallow	none
Knowledge Graph	first-class, queryable edges	flexible, evolving ontology	automatic reasoning

- **Relational DB:** superb for stable, tabular data — but cross-source clinical queries become **tangled JOINS**, and the schema cannot express *“if X then Y”*.
- **Document store:** easy to ingest heterogeneous blobs — but there are **no explicit relationships and no reasoning**; each document is an island.
- **KG:** relationships are **the data**; the ontology can evolve; and a reasoner **derives consequences** the database would never surface.

i Discussion Prompt for PhD Students

3 · Clinical Relationships — Complexity & Patterns

5 minutes — “*Not all edges are created equal.*”

- A KG's power lives in its **edges** — but clinical relationships obey **structural patterns** that the schema must respect.
- Modelling the *wrong pattern* produces the *wrong inference* — a clinical safety issue, not a cosmetic one.

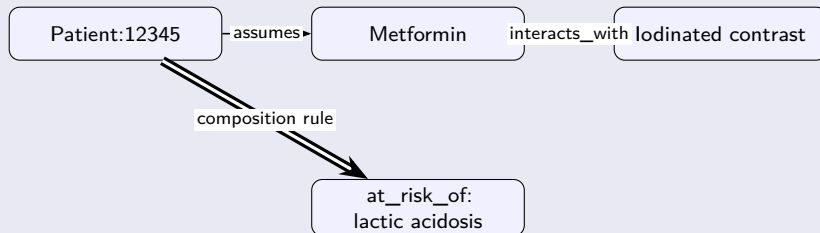
i Discussion Prompt for PhD Students

If your KG silently treated causes as symmetric, what specific clinical error could it commit? Find one before the next slide.

Pattern	Healthcare example	Property
Symmetric	sibling_of (rare clinically)	if $A \sim B$ then $B \sim A$
Antisymmetric	parent_of, causes	$A \rightarrow B$ forbids $B \rightarrow A$
Inverse	prescribes prescribed_by; monitors monitored_by	two names, one fact
Composition	assumes interacts_with $\rightarrow$ at_risk_of_interaction	new edge from two
One-to-many	one patient $\rightarrow$ many diagnoses, meds, observations	high fan-out

- **Antisymmetry** matters: if *A causes B*, then *B does not cause A* — confusing this inverts aetiology.
- **Inverse pairs** let you query from either end without duplicating data (“*which drugs does this patient take*” “*which patients take this drug*”).
- **Composition** is how **reasoning creates new knowledge**: two traversed edges yield a third edge that was never stored.





- Two stored edges — `assumes` and `interacts_with` — are **composed** into a derived edge `at_risk_of`.
- The derived edge **never existed in the raw data**: it is the KG *thinking*, not just storing.
- **One-to-many** fan-out amplifies this: a patient with many medications triggers many composition checks in parallel — exactly the workload that defeats manual clinical review.

! Why this section matters

If the schema mis-models a relationship's pattern (say, `treats` causes as symmetric, or omits an inverse), **reasoning silently degrades**. Correct relationship modelling is the precondition for trustworthy clinical inference.

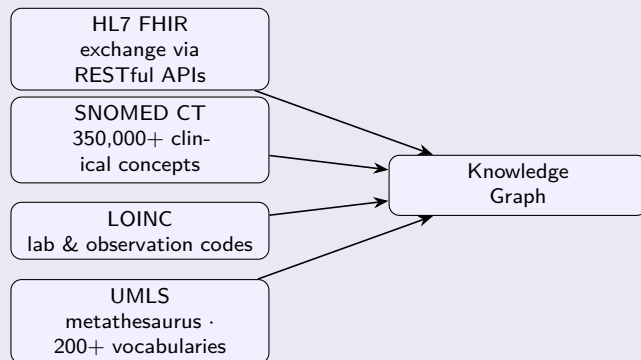
4 · Healthcare Ontologies — The Controlled Vocabulary

2 minutes — *“You don’t invent the vocabulary.”*

- A KG is only as interoperable as the **codes on its nodes**. In healthcare, that means **reusing standard ontologies**, not minting your own.

i Discussion Prompt for PhD Students

You could, in principle, invent a private vocabulary for your hospital’s KG. Give one concrete reason a regulator would reject it — and one reason your own researchers would resent it.



- **HL7 FHIR** — the *exchange* spine: healthcare data over **RESTful APIs** (resources, not just codes).
- **SNOMED CT** — the *clinical terminology*: **350,000+ concepts** covering diagnoses, procedures, findings.
- **LOINC** — *lab & observation* coding (e.g., the glucose code 99504-3 on our mini-KG).
- **UMLS** — a *metathesaurus* that cross-maps **200+ vocabularies** so they can speak to each other.



Why you cannot roll your own ontology

Key Takeaways

- Healthcare data is heterogeneous on **four independent axes** (structural, semantic, temporal, spatial) — volume is the easy part.
- A **knowledge graph** is a directed, labelled, semantically grounded graph built on **three pillars**: data, schema, and inference (Hogan et al., 2021).
- The KG **absorbs heterogeneity** by standardising codes (SNOMED CT, LOINC), linking sources, and **deriving** clinically consequential facts (e.g., lactic-acidosis risk).
- **Correct relationship modelling** (symmetric / antisymmetric / inverse / composition) is the precondition for **safe** clinical reasoning.
- **Reuse standard ontologies** (FHIR, SNOMED CT, LOINC, UMLS) — interoperability, reuse, and regulation forbid inventing your own.

i Discussion Prompt for PhD Students — revisit

Return to the integration failure you named on slide 1. Which of the four heterogeneity dimensions, and which missing KG pillar, best explains it? Has your diagnosis changed in 25 minutes?

i Discussion Prompt for PhD Students

These sources span a formal survey, a standard, and a clinical case study. Which *type* of source is most under-represented for your thesis — and where would you look for it?

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- Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., de Melo, G., Gutierrez, C., et al. (2021). Knowledge Graphs. *ACM Computing Surveys*, 54(4), Article 71. <https://doi.org/10.1145/3447772>
- Leskovec, J. (2021). *CS224W: Machine Learning with Graphs — Knowledge Graphs* [Lecture slides, slides 6–9 and 24–28]. Stanford University. <https://web.stanford.edu/class/cs224w/>
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- SNOMED International. (n.d.). *SNOMED CT*. <https://www.snomed.org/>